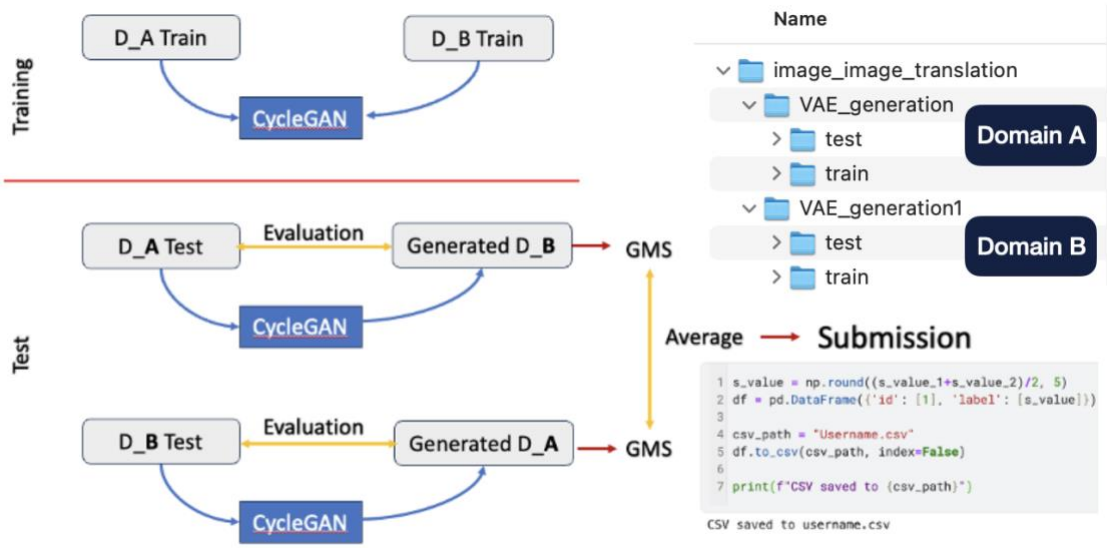
CS424 Group Project: CycleGAN for Image-to-Image Translation

(Friday-Class)

Task Overview: In this project, you will gain hands-on experience in coding and training CycleGAN for image-to-image translation, focusing on translating images from domain A to domain B and vice versa. Given Domain A Cat and Domain B Dog (Unpaired) data, we learn a CycleGAN model, which can translate an image from Domain A to Domain B or Domain B to Domain A.



This assignment consists of two tasks:

- **Task 1 (Group but needs individual submission on Kaggle)**: Train a CycleGAN on the provided dataset to translate images between **Domain A** and **Domain B**.
- **Task 2 (No individual submission on Kaggle)**: Apply CycleGAN to another image-to-image translation task involving two different domains. Note: Using CycleGAN on new datasets is also considered novel. Tasks/datasets already covered in the CycleGAN paper will score lower on novelty.
- **Submission:**
  - Kaggle: **Each group member** needs to submit the same result of task 1 on Kaggle. Please do not submit any results of task 2 on Kaggle.
  - eLearn: Team leader in each group submits one zip file (including pdf report and code for task 1 and task 2, pdf presentation slide) on eLearn.

You are encouraged to enhance the vanilla CycleGAN for both tasks by:

1. Designing new neural network architectures or leveraging existing network.
2. Improving the training loss function.

3. Enhancing optimization strategies, such as advanced optimizers or hyperparameter tuning (e.g., learning rate adjustments).

Important: All enhancements must adhere to the specified requirements. Failure to comply will result in a score of zero.

Requirements:

1. **Data Usage:**
   - **Task 1**: Use only the provided training dataset. Conventional data augmentation is allowed, but using external data or pre-trained models (like generative models) is strictly prohibited. Download the training data here.
   - **Task 2**: No restrictions. You may use existing datasets or collect new data for training and testing.
2. **Baseline Code:**
   - **Task 1**: Use the provided test functions to evaluate generation performance after training. **Do not alter the provided test data and the hyperparameters and code in the test function** colab.
   - **Task 2**: You may use the provided test functions or other metrics to evaluate performance. This is open. The key is to select suitable metrics for evaluation.
3. **Training Constraints:**
   - **Task 1**: Models must be trained from scratch, starting with random weight initialization. Pre-trained models are not allowed. You may improve the vanilla CycleGAN training loss but cannot incorporate losses from other generative models (e.g., diffusion models).
   - **Task 2**: No restrictions on training strategies or techniques. You may improve the vanilla CycleGAN training loss but cannot incorporate losses from other generative models (e.g., diffusion models).
4. **Model Size:**
   - **Task 1**: The model's total parameter size must not exceed 60M. Use the provided function to verify this.
   - **Task 2**: No restrictions.
5. **Report Page Limit:**
   - The report must not exceed **12 pages**.
6. **Submission Deadline:**
   - Submit your result for Task 1 on Kaggle kaggle by **April 9, 2026, 11:55 PM**. Please do not submit the result of Task 2, since group tasks are different and the results are not comparable.
   - Submit your work (pdf report and code of task 1 and 2, and pdf presentation slide) on eLearn by **April 9, 2026, 11:59 PM**. Team leader in each group submits one zip file (including pdf report and code for task 1 and task 2) on eLearn.
   - Oral Presentation: during class in Week 13 (15 minutes for presentation + 5 minutes for QA session)
7. **Consistency between Kaggle score and Report in eLearn**
   - Please ensure that **for task 1, your Kaggle score is consistent with your eLearn**

submission. If there is a discrepancy, please contact the TA immediately. Unresolved inconsistencies will result in **a 5% penalty** to the performance component of your grade, as we cannot produce the Kaggle score.

Submission Instructions:

1. **Group submission**: Each team should register in the Excel excel. Team leader should submit the following in a single ZIP file (with team name, e.g., "T1.zip", "T2.zip") via eLearn:

- **Report**: A PDF document.
- **Code**: All relevant code files.
- **Presentation slides (pdf version)**

2. **Individual submission**: each team member should submit your result for Task 1 on Kaggle kaggle.

Evaluation Criteria:

Task 1 (40%)

1. **Performance (60%)**:
   - Performance will be evaluated based on the results from the provided test functions. Higher performance scores will yield higher grades.
2. **Report Quality (35%)**:
   - **Basic Introduction (15%)**: Provide detailed descriptions of key components, including training loss, network architecture, model size, and performance.
   - **Improvements (10%)**: Highlight innovative designs or enhancements, such as network modifications or improved training loss functions.
   - **Findings/Insights (10%)**: Discuss key observations and insights gained during model training.
3. **Code Quality (5%)**:
   - Assessed based on clarity, organization, and adherence to best coding practices.

Task 2 (30%)

1. **Task Novelty and Challenge (30%)**:
   - Select a novel or interesting image-to-image translation task. Explain why the task is novel or challenging. Note: Using CycleGAN on new datasets is also considered novel. Tasks/datasets already covered in the CycleGAN paper will score lower on novelty.
   - Higher difficulty tasks requiring advanced techniques or significant effort will score higher.
2. **Performance (30%)**:

    - Scores will depend on generation performance evaluated using the provided test functions or other metrics.
3. **Report Quality (35%)**:
    - **Basic Introduction (15%)**: Describe all components, including training loss, network architecture, model size, and performance.
    - **Improvements (10%)**: Highlight innovative designs or modifications made to improve performance.
    - **Findings/Insights (10%)**: Provide key observations and insights from the training process.
4. **Code Quality (5%)**:
    - Assessed based on clarity, organization, and adherence to coding best practices.

Oral Presentation (30%)

This part consists of 15-minute presentation and 5-minute QA session. Each team will be evaluated by both the instructor and other teams based on the following aspects. The instructor will determine 20% of the score, while the remaining 10% will be assigned based on peer evaluations from other teams.

- **Slides (30%)**: Assessed for clarity, organization, and readability.
- **Presentation (30%)**: Evaluated on clarity, structure, and engagement.
- **Technical Depth and Novelty (40%)**: Assessed based on the task's novelty, level of challenge, and the use of advanced techniques. This part also includes the performance of QA session.